

9D Centralized Maximum Likelihood Kalman Filter

I. SYSTEM STATE AND KINEMATICS

Consider a cooperative localization system consisting of I nodes, indexed by $i \in \{1, \dots, I\}$. Let the global frame be the navigation frame (n -frame) and the local body-fixed frame be the b -frame. At discrete time step t , the state of node i is defined on the manifold $\mathcal{M} = \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3)$:

$$\mathbf{x}_t^i = \begin{bmatrix} \mathbf{p}_t^i \\ \mathbf{v}_t^i \\ \mathbf{R}_t^i \end{bmatrix} \in \mathcal{M} \quad (1)$$

where $\mathbf{p}_t^i \in \mathbb{R}^3$ denotes position in the n -frame, $\mathbf{v}_t^i \in \mathbb{R}^3$ denotes linear velocity in the n -frame, and $\mathbf{R}_t^i = \mathbf{R}_{nb,t}^i \in SO(3)$ is the rotation matrix transforming coordinates from the b -frame to the n -frame. The joint state of the multi-vehicle system is stacked as:

$$\mathbf{x}_t = [(\mathbf{x}_t^1)^\top, \dots, (\mathbf{x}_t^I)^\top]^\top \in \mathcal{M}^I \quad (2)$$

The IMU measurements of vehicle i consist of specific force $\check{\mathbf{a}}_t^i \in \mathbb{R}^3$ and angular velocity $\check{\boldsymbol{\omega}}_t^i \in \mathbb{R}^3$ resolved in the b -frame:

$$\check{\mathbf{a}}_t^i = (\mathbf{R}_t^i)^\top (\mathbf{a}_t^i - \mathbf{g}) + \bar{\mathbf{a}}^i + \mathbf{e}_{a,t}^i \quad (3)$$

$$\check{\boldsymbol{\omega}}_t^i = \boldsymbol{\omega}_t^i + \bar{\boldsymbol{\omega}}^i + \mathbf{e}_{\omega,t}^i \quad (4)$$

where $\mathbf{a}_t^i \in \mathbb{R}^3$ is true acceleration in the n -frame, $\mathbf{g} \in \mathbb{R}^3$ is the gravity vector in the n -frame, $\bar{\mathbf{a}}^i, \bar{\boldsymbol{\omega}}^i$ are known constant sensor biases, and the measurement noises are Gaussian white sequences:

$$\mathbf{e}_{a,t}^i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_a^i), \quad \mathbf{e}_{\omega,t}^i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\omega^i) \quad (5)$$

The deterministic control inputs driving the kinematic propagation are defined via bias-corrected IMU signals:

$$\mathbf{a}_{m,t}^i \triangleq \check{\mathbf{a}}_t^i - \bar{\mathbf{a}}^i, \quad \boldsymbol{\omega}_{m,t}^i \triangleq \check{\boldsymbol{\omega}}_t^i - \bar{\boldsymbol{\omega}}^i \quad (6)$$

II. LIE GROUP AND MANIFOLD OPERATIONS

The Lie algebra $\mathfrak{so}(3)$ is associated with 3×3 skew-symmetric matrices via the map $[\cdot]_\times : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$. The exponential map $\exp : \mathbb{R}^3 \rightarrow SO(3)$ is given by Rodrigues' formula:

$$\exp([\phi]_\times) = \mathbf{I}_3 + \frac{\sin \theta}{\theta} [\phi]_\times + \frac{1 - \cos \theta}{\theta^2} [\phi]_\times^2, \quad \theta = \|\phi\| \quad (7)$$

The right Jacobian of $SO(3)$, denoted as $J_r(\phi)$, satisfies:

$$J_r(\phi) = \mathbf{I}_3 - \frac{1 - \cos \theta}{\theta^2} [\phi]_\times + \frac{\theta - \sin \theta}{\theta^3} [\phi]_\times^2 \quad (8)$$

III. PREDICTION

A. Nominal State Propagation

Given the posterior state estimate $\hat{\mathbf{x}}_{t|t}^i$ at time step t and sampling period τ , the prior state estimate $\hat{\mathbf{x}}_{t+1|t}^i$ propagates via numerical integration:

$$\hat{\mathbf{R}}_{t+1|t}^i = \hat{\mathbf{R}}_{t|t}^i \exp([\tau \boldsymbol{\omega}_{m,t}^i]_\times) \quad (9)$$

$$\hat{\mathbf{v}}_{t+1|t}^i = \hat{\mathbf{v}}_{t|t}^i + \tau (\hat{\mathbf{R}}_{t|t}^i \mathbf{a}_{m,t}^i + \mathbf{g}) \quad (10)$$

$$\hat{\mathbf{p}}_{t+1|t}^i = \hat{\mathbf{p}}_{t|t}^i + \tau \hat{\mathbf{v}}_{t|t}^i + \frac{\tau^2}{2} (\hat{\mathbf{R}}_{t|t}^i \mathbf{a}_{m,t}^i + \mathbf{g}) \quad (11)$$

B. Error State Dynamics Derivation

Let the true state at t and $t+1$ be parameterized by the posterior error $\boldsymbol{\theta}_{t|t}^i$ and prior error $\boldsymbol{\theta}_{t+1|t}^i$, respectively:

$$\mathbf{R}_t^i = \hat{\mathbf{R}}_{t|t}^i \exp([\delta \phi_{t|t}^i]_\times) \quad (12)$$

$$\mathbf{R}_{t+1}^i = \hat{\mathbf{R}}_{t+1|t}^i \exp([\delta \phi_{t+1|t}^i]_\times) \quad (13)$$

1) *Attitude Error Propagation:* The true attitude evolves according to:

$$\mathbf{R}_{t+1}^i = \mathbf{R}_t^i \exp([\tau(\boldsymbol{\omega}_{m,t}^i - \mathbf{e}_{\omega,t}^i)]_\times) \quad (14)$$

Substituting (12) and (13) into the physical propagation yields:

$$\begin{aligned} \hat{\mathbf{R}}_{t+1|t}^i \exp([\delta \phi_{t+1|t}^i]_\times) &= \\ \hat{\mathbf{R}}_{t|t}^i \exp([\delta \phi_{t|t}^i]_\times) \exp([\tau(\boldsymbol{\omega}_{m,t}^i - \mathbf{e}_{\omega,t}^i)]_\times) \end{aligned} \quad (15)$$

Multiplying both sides from the left by $(\hat{\mathbf{R}}_{t+1|t}^i)^\top$:

$$\begin{aligned} \exp([\delta \phi_{t+1|t}^i]_\times) &= (\hat{\mathbf{R}}_{t+1|t}^i)^\top \hat{\mathbf{R}}_{t|t}^i \exp([\delta \phi_{t|t}^i]_\times) \\ &\quad \cdot \exp([\tau(\boldsymbol{\omega}_{m,t}^i - \mathbf{e}_{\omega,t}^i)]_\times) \end{aligned} \quad (16)$$

From the nominal equation (9), we evaluate:

$$(\hat{\mathbf{R}}_{t+1|t}^i)^\top \hat{\mathbf{R}}_{t|t}^i = \exp(-\tau[\boldsymbol{\omega}_{m,t}^i]_\times) (\hat{\mathbf{R}}_{t|t}^i)^\top \hat{\mathbf{R}}_{t|t}^i = \exp(-\tau[\boldsymbol{\omega}_{m,t}^i]_\times) \quad (17)$$

Applying the first-order right Jacobian expansion to the noise perturbation:

$$\exp([\tau(\boldsymbol{\omega}_{m,t}^i - \mathbf{e}_{\omega,t}^i)]_\times) \approx \exp(\tau[\boldsymbol{\omega}_{m,t}^i]_\times) \exp([\tau J_r(\tau \boldsymbol{\omega}_{m,t}^i) \mathbf{e}_{\omega,t}^i]_\times) \quad (18)$$

Using the adjoint identity $\exp(\mathbf{u}_\times) \exp(\mathbf{v}_\times) \exp(-\mathbf{u}_\times) = \exp([\exp(\mathbf{u}_\times) \mathbf{v}]_\times)$, the error equation expands to first-order:

$$\delta \phi_{t+1|t}^i \approx \exp(-\tau[\boldsymbol{\omega}_{m,t}^i]_\times) \delta \phi_{t|t}^i - \tau J_r(\tau \boldsymbol{\omega}_{m,t}^i) \mathbf{e}_{\omega,t}^i \quad (19)$$

2) *Velocity Error Propagation*: The true velocity propagation satisfies:

$$\mathbf{v}_{t+1}^i = \mathbf{v}_t^i + \tau (\mathbf{R}_t^i (\mathbf{a}_{m,t}^i - \mathbf{e}_{a,t}^i) + \mathbf{g}) \quad (20)$$

Substituting $\mathbf{R}_t^i = \hat{\mathbf{R}}_{t|t}^i \exp([\delta\phi_{t|t}^i]_{\times}) \approx \hat{\mathbf{R}}_{t|t}^i (\mathbf{I}_3 + [\delta\phi_{t|t}^i]_{\times})$:

$$\begin{aligned} \mathbf{v}_{t+1}^i &\approx \hat{\mathbf{v}}_{t|t}^i + \delta\mathbf{v}_{t|t}^i + \tau \hat{\mathbf{R}}_{t|t}^i \mathbf{a}_{m,t}^i + \tau \mathbf{g} \\ &\quad + \tau \hat{\mathbf{R}}_{t|t}^i [\delta\phi_{t|t}^i]_{\times} \mathbf{a}_{m,t}^i - \tau \hat{\mathbf{R}}_{t|t}^i \mathbf{e}_{a,t}^i \end{aligned} \quad (21)$$

Using the skew-symmetric identity $[\mathbf{u}]_{\times} \mathbf{v} = -[\mathbf{v}]_{\times} \mathbf{u}$:

$$\hat{\mathbf{R}}_{t|t}^i [\delta\phi_{t|t}^i]_{\times} \mathbf{a}_{m,t}^i = -\hat{\mathbf{R}}_{t|t}^i [\mathbf{a}_{m,t}^i]_{\times} \delta\phi_{t|t}^i \quad (22)$$

Subtracting the nominal propagation (10) from the true expression yields:

$$\delta\mathbf{v}_{t+1|t}^i = \delta\mathbf{v}_{t|t}^i - \tau \hat{\mathbf{R}}_{t|t}^i [\mathbf{a}_{m,t}^i]_{\times} \delta\phi_{t|t}^i - \tau \hat{\mathbf{R}}_{t|t}^i \mathbf{e}_{a,t}^i \quad (23)$$

3) *Position Error Propagation*: Integrating the kinematic relation for position over period τ :

$$\mathbf{p}_{t+1}^i = \mathbf{p}_t^i + \tau \mathbf{v}_t^i + \frac{\tau^2}{2} (\mathbf{R}_t^i (\mathbf{a}_{m,t}^i - \mathbf{e}_{a,t}^i) + \mathbf{g}) \quad (24)$$

Subtracting the nominal propagation (11) and retaining first-order terms:

$$\delta\mathbf{p}_{t+1|t}^i = \delta\mathbf{p}_{t|t}^i + \tau \delta\mathbf{v}_{t|t}^i - \frac{\tau^2}{2} \hat{\mathbf{R}}_{t|t}^i [\mathbf{a}_{m,t}^i]_{\times} \delta\phi_{t|t}^i - \frac{\tau^2}{2} \hat{\mathbf{R}}_{t|t}^i \mathbf{e}_{a,t}^i \quad (25)$$

C. Linearized State Space Model and Covariance Propagation

Combining (25), (23), and (19), the discrete-time linear error dynamics for node i are:

$$\boldsymbol{\theta}_{t+1|t}^i = \mathbf{A}_t^i \boldsymbol{\theta}_{t|t}^i + \mathbf{W}_t^i \mathbf{w}_t^i \quad (26)$$

where the input noise vector is $\mathbf{w}_t^i = [(\mathbf{e}_{a,t}^i)^\top, (\mathbf{e}_{\omega,t}^i)^\top]^\top$, and the local transition and input matrices are:

$$\mathbf{A}_t^i = \begin{bmatrix} \mathbf{I}_3 & \tau \mathbf{I}_3 & -\frac{\tau^2}{2} \hat{\mathbf{R}}_{t|t}^i [\mathbf{a}_{m,t}^i]_{\times} \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_3 & -\tau \hat{\mathbf{R}}_{t|t}^i [\mathbf{a}_{m,t}^i]_{\times} \\ \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times 3} & \exp(-\tau [\boldsymbol{\omega}_{m,t}^i]_{\times}) \end{bmatrix} \quad (27)$$

$$\mathbf{W}_t^i = \begin{bmatrix} -\frac{\tau^2}{2} \hat{\mathbf{R}}_{t|t}^i & \mathbf{0}_{3 \times 3} \\ -\tau \hat{\mathbf{R}}_{t|t}^i & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & -\tau J_r(\tau \boldsymbol{\omega}_{m,t}^i) \end{bmatrix} \quad (28)$$

The discrete local process noise covariance matrix $\mathbf{Q}_t^i \in \mathbb{R}^{9 \times 9}$ is:

$$\mathbf{Q}_t^i = \mathbf{W}_t^i \begin{bmatrix} \boldsymbol{\Sigma}_a^i & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \boldsymbol{\Sigma}_\omega^i \end{bmatrix} (\mathbf{W}_t^i)^\top \quad (29)$$

For the centralized multi-vehicle network of I nodes, the global transition and process noise matrices are block-diagonal:

$$\mathbf{A}_t = \text{blockdiag}(\mathbf{A}_t^1, \dots, \mathbf{A}_t^I) \in \mathbb{R}^{9I \times 9I} \quad (30)$$

$$\mathbf{Q}_t = \text{blockdiag}(\mathbf{Q}_t^1, \dots, \mathbf{Q}_t^I) \in \mathbb{R}^{9I \times 9I} \quad (31)$$

The global error covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{9I \times 9I}$ propagates via:

$$\boldsymbol{\Sigma}_{t+1|t} = \mathbf{A}_t \boldsymbol{\Sigma}_{t|t} \mathbf{A}_t^\top + \mathbf{Q}_t \quad (32)$$

IV. UPDATE AND FUSION

A. Measurement Modeling and Subspace Projection

At time step $t+1$, the measurement vector $\mathbf{y}_{t+1}$ comprises UWB range observations between nodes and known anchors $\mathbf{c}^k \in \mathbb{R}^3$ ($k \in \mathcal{K}$), as well as inter-node ranges between node i and neighbor $j \in \mathcal{N}_i$:

$$s_{t+1}^{i,k} = \|\mathbf{p}_{t+1}^i - \mathbf{c}^k\| + e_{s,t+1}^{i,k}, \quad e_{s,t+1}^{i,k} \sim \mathcal{N}(0, \sigma_s^2) \quad (33)$$

$$z_{t+1}^{i,j} = \|\mathbf{p}_{t+1}^i - \mathbf{p}_{t+1}^j\| + e_{z,t+1}^{i,j}, \quad e_{z,t+1}^{i,j} \sim \mathcal{N}(0, \sigma_z^2) \quad (34)$$

Stacking all available measurements yields the general observation equation:

$$\mathbf{y}_{t+1} = h(\mathbf{p}_{t+1}) + \mathbf{v}_{t+1}, \quad \mathbf{v}_{t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{\text{UWB}}) \quad (35)$$

where $\mathbf{R}_{\text{UWB}} \in \mathbb{R}^{M \times M}$ is the diagonal measurement noise covariance matrix.

Because velocity and attitude are not directly observable from range measurements, we define the state selection operator $\boldsymbol{\Pi} : \mathcal{M}^I \rightarrow \mathbb{R}^{3I}$ isolating the position components:

$$\boldsymbol{\chi}_{t+1} = \boldsymbol{\Pi} \mathbf{x}_{t+1} = [(\mathbf{p}_{t+1}^1)^\top, \dots, (\mathbf{p}_{t+1}^I)^\top]^\top \in \mathbb{R}^{3I} \quad (36)$$

In the tangent space, the linear selection matrix $\boldsymbol{\pi} \in \mathbb{R}^{3I \times 9I}$ projects the full-dimensional error state $\boldsymbol{\theta}_{t+1} \in \mathbb{R}^{9I}$ onto the measurement-relevant error vector $\mathbf{s}_{t+1} \in \mathbb{R}^{3I}$:

$$\mathbf{s}_{t+1} = \boldsymbol{\pi} \boldsymbol{\theta}_{t+1}, \quad \boldsymbol{\pi} = \text{blockdiag}(\boldsymbol{\pi}^1, \dots, \boldsymbol{\pi}^I) \quad (37)$$

where each local block is defined by:

$$\boldsymbol{\pi}^i = [\mathbf{I}_3 \quad \mathbf{0}_{3 \times 3} \quad \mathbf{0}_{3 \times 3}] \in \mathbb{R}^{3 \times 9} \quad (38)$$

B. Iterative Maximum Likelihood Optimization

The Maximum Likelihood (ML) estimation is formulated as a non-linear least-squares problem over the Euclidean position manifold:

$$\min_{\mathbf{p}} \mathcal{J}(\mathbf{p}) = \frac{1}{2} \|\mathbf{y}_{t+1} - h(\mathbf{p})\|_{\mathbf{R}_{\text{UWB}}^{-1}}^2 \quad (39)$$

To solve (39) via the Gauss-Newton method, let $\mathbf{p}^{(l)} \in \mathbb{R}^{3I}$ denote the state estimate at iteration l , initialized by the prior prediction $\mathbf{p}^{(0)} = \hat{\mathbf{p}}_{t+1|t}$.

Applying a local Euclidean perturbation $\delta\mathbf{p} = [(\delta\mathbf{p}^1)^\top, \dots, (\delta\mathbf{p}^I)^\top]^\top \in \mathbb{R}^{3I}$, the measurement Jacobian $\mathbf{H}^{(l)} \in \mathbb{R}^{M \times 3I}$ is evaluated at $\mathbf{p}^{(l)}$:

$$\mathbf{H}^{(l)} = \left. \frac{\partial h(\mathbf{p}^{(l)} + \delta\mathbf{p})}{\partial \delta\mathbf{p}} \right|_{\delta\mathbf{p}=\mathbf{0}} \quad (40)$$

The analytical non-zero Jacobian blocks for the measurement models are:

$$\frac{\partial s_{t+1}^{i,k}}{\partial \mathbf{p}^i} = (\mathbf{u}_{t+1}^{i,k})^\top \quad (41)$$

$$\frac{\partial z_{t+1}^{i,j}}{\partial \mathbf{p}^i} = (\mathbf{u}_{t+1}^{i,j})^\top, \quad \frac{\partial z_{t+1}^{i,j}}{\partial \mathbf{p}^j} = -(\mathbf{u}_{t+1}^{i,j})^\top, \quad (42)$$

$$\mathbf{u}_{t+1}^{i,j} \triangleq \frac{\mathbf{p}^{i,(l)} - \mathbf{p}^{j,(l)}}{\|\mathbf{p}^{i,(l)} - \mathbf{p}^{j,(l)}\|} \quad (43)$$

At iteration l , with measurement residual $\mathbf{r}^{(l)} = \mathbf{y}_{t+1} - h(\mathbf{p}^{(l)})$, the correction increment $\Delta \mathbf{p}^{(l)} \in \mathbb{R}^{3I}$ is computed by solving the normal equations (Ensure reversibility):

$$\left((\mathbf{H}^{(l)})^\top \mathbf{R}_{\text{UWB}}^{-1} \mathbf{H}^{(l)} \right) \Delta \mathbf{p}^{(l)} = (\mathbf{H}^{(l)})^\top \mathbf{R}_{\text{UWB}}^{-1} \mathbf{r}^{(l)} \quad (44)$$

The position estimate updates additively:

$$\mathbf{p}^{(l+1)} = \mathbf{p}^{(l)} + \Delta \mathbf{p}^{(l)} \quad (45)$$

C. Likelihood Information Extraction

Upon convergence at iteration L satisfying $\|\Delta \mathbf{p}^{(L)}\| < \epsilon$, the ML state error vector $\boldsymbol{\mu}_{t+1} \in \mathbb{R}^{3I}$ relative to the prior estimate $\hat{\mathbf{p}}_{t+1|t}$ and its associated Fisher Information Matrix $\boldsymbol{\Xi}_{t+1}^{-1} \in \mathbb{R}^{3I \times 3I}$ are extracted:

$$\boldsymbol{\mu}_{t+1} = \mathbf{p}^{(L)} - \hat{\mathbf{p}}_{t+1|t} \quad (46)$$

$$\boldsymbol{\Xi}_{t+1}^{-1} = (\mathbf{H}^{(L)})^\top \mathbf{R}_{\text{UWB}}^{-1} \mathbf{H}^{(L)} \quad (47)$$

D. Prior and Likelihood Fusion

Lifting the likelihood information into the full $9I$ -dimensional error space via the selection matrix $\boldsymbol{\pi}$ yields the global measurement information matrix $\boldsymbol{\Lambda}_{t+1} \in \mathbb{R}^{9I \times 9I}$ and information vector $\boldsymbol{\lambda}_{t+1} \in \mathbb{R}^{9I}$:

$$\boldsymbol{\Lambda}_{t+1} = \boldsymbol{\pi}^\top \boldsymbol{\Xi}_{t+1}^{-1} \boldsymbol{\pi} \quad (48)$$

$$\boldsymbol{\lambda}_{t+1} = \boldsymbol{\pi}^\top \boldsymbol{\Xi}_{t+1}^{-1} \boldsymbol{\mu}_{t+1} \quad (49)$$

The posterior error covariance matrix $\boldsymbol{\Sigma}_{t+1|t+1} \in \mathbb{R}^{9I \times 9I}$ is updated in information form:

$$\boldsymbol{\Sigma}_{t+1|t+1} = \left(\boldsymbol{\Sigma}_{t+1|t}^{-1} + \boldsymbol{\Lambda}_{t+1} \right)^{-1} \quad (50)$$

Alternatively, applying the Woodbury matrix identity to (50) yields the equivalent computationally efficient form:

$$\mathbf{K}_{t+1} = \boldsymbol{\Sigma}_{t+1|t} \boldsymbol{\pi}^\top \left(\boldsymbol{\pi} \boldsymbol{\Sigma}_{t+1|t} \boldsymbol{\pi}^\top + \boldsymbol{\Xi}_{t+1} \right)^{-1} \quad (51)$$

$$\boldsymbol{\Sigma}_{t+1|t+1} = (\mathbf{I}_{9I} - \mathbf{K}_{t+1} \boldsymbol{\pi}) \boldsymbol{\Sigma}_{t+1|t} \quad (52)$$

The full-dimensional state correction vector $\Delta \boldsymbol{\theta}_{t+1} \in \mathbb{R}^{9I}$ is computed as:

$$\Delta \boldsymbol{\theta}_{t+1} = \boldsymbol{\Sigma}_{t+1|t+1} \boldsymbol{\lambda}_{t+1} \quad (53)$$

For each node i , parsing $\Delta \boldsymbol{\theta}_{t+1}^i = [(\delta \mathbf{p}^i)^\top, (\delta \mathbf{v}^i)^\top, (\delta \boldsymbol{\phi}^i)^\top]^\top \in \mathbb{R}^9$, the posterior state estimate is mapped onto $\mathcal{M}$ via the retraction operator:

$$\hat{\mathbf{p}}_{t+1|t+1}^i = \hat{\mathbf{p}}_{t+1|t}^i + \delta \mathbf{p}^i \quad (54)$$

$$\hat{\mathbf{v}}_{t+1|t+1}^i = \hat{\mathbf{v}}_{t+1|t}^i + \delta \mathbf{v}^i \quad (55)$$

$$\hat{\mathbf{R}}_{t+1|t+1}^i = \hat{\mathbf{R}}_{t+1|t}^i \exp([\delta \boldsymbol{\phi}^i]_\times) \quad (56)$$